

Project Initialization and Planning Phase

Date	03 October 2026
Team ID	SWTID-2026-9455
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

ComicCraft applies generative AI to comic creation. From a short prompt and a few preferences it writes a 5-panel story, draws every panel and exports a PDF, making comic making accessible to non-artists.

Project Overview

Objective	Automate comic creation end to end: personalised storyline, dialogue, illustrations and a downloadable PDF from one simple web form.
Scope	Web app (FastAPI + Jinja2) with a JSON API; text with Gemini Flash and Pro, images with Stable Diffusion, PDF with FPDF2; runs locally. Accounts, storage of past comics and cloud hosting are out of scope.

Problem Statement

Description	People with story ideas but no drawing skills, time or budget cannot turn them into finished, illustrated comics, and existing tools split writing, art and layout.
Impact	Solving this lets students, teachers, parents and creators produce shareable comics in minutes, supports learning and creativity, and lowers the barrier to visual storytelling.

Proposed Solution

Approach	A staged AI pipeline: Gemini Flash creates a structured 5-panel outline, Gemini Pro expands it into narration and dialogue, Stable Diffusion renders each panel in the chosen art style, and a layout builder plus FPDF2 compile the PDF.
Key Features	• Personalisation: story prompt, character name, setting, tone, art style • Two-model text pipeline (fast outline, rich story) • Comic-style image per panel • On-screen preview, PDF download and success page • JSON API and image test route • Validation, retries, error handling and offline mock mode

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	NVIDIA T4 (16 GB) GPU recommended for Stable Diffusion (free on Colab/Kaggle); CPU works but is slow
Memory	RAM specifications	8 GB minimum, 16 GB recommended
Storage	Disk space for models, images and PDFs	About 10 GB free (Stable Diffusion weights are several GB) plus generated files
Software		
Frameworks	Python frameworks	FastAPI, Uvicorn, Jinja2

Resource Type	Description	Specification / Allocation
Libraries	Additional libraries	google-genai, diffusers, transformers, torch, accelerate, fpdf2, Pillow, python-dotenv, pydantic, pytest, httpx
Development Environment	IDE	VS Code or PyCharm, Python 3.10+, Git
Data / APIs	Source, size, format	No training dataset (pre-trained models). Gemini API key (Google AI Studio) and Hugging Face access token